WORKED SOLUTION

STEP II 1996, Question 14

Probability and Statistics • Moments and Central Limit Theorem

STEP II 1996, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q14

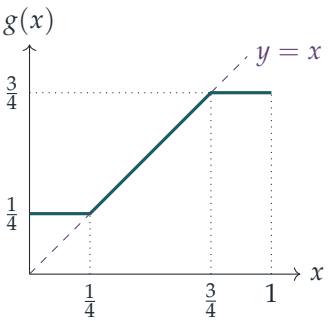
The key idea

Y is X with its two ends "clipped": all the probability below $\frac{1}{4}$ is piled up at the single value $\frac{1}{4}$, and all the probability above $\frac{3}{4}$ at $\frac{3}{4}$. So Y is a mixed random variable, with two point masses and a continuous middle piece, and any expectation $\mathrm{E}(g(Y))$ splits into three pieces in the same way. The final part has a sum of a very large number of independent copies, which calls for the central limit theorem. For that we need only the mean and variance of Y, and the earlier parts have already done that work.

Solution

The moments of Y

Since $X \sim \mathrm{U}[0,1]$, we have $\mathrm{P}(X \leq \frac{1}{4}) = \mathrm{P}(X \geq \frac{3}{4}) = \frac{1}{4}$, and on $[\frac{1}{4}, \frac{3}{4}]$ the density of X is 1. Writing $Y = g(X)$, where g is the clipping function sketched below, we get $\mathrm{E}(Y^n) = \int_0^1 g(x)^n \, \mathrm{d}x$.



Splitting the integral at $\frac{1}{4}$ and $\frac{3}{4}$,

$$\begin{aligned}\mathrm{E}(Y^n) &= \left(\tfrac{1}{4}\right)^n \cdot \tfrac{1}{4} + \int_{1/4}^{3/4} x^n \,\mathrm{d}x + \left(\tfrac{3}{4}\right)^n \cdot \tfrac{1}{4} \\ &= \frac{1}{4^{n+1}} + \frac{3^{n+1}-1}{(n+1)4^{n+1}} + \frac{3^n}{4^{n+1}} \\ &= \frac{(n+1)(1+3^n)+3^{n+1}-1}{(n+1)4^{n+1}}.\end{aligned}$$

The numerator is $n+1+(n+1)3^n+3\cdot 3^n-1 = n+(n+4)3^n$, so

$$\boxed{\mathrm{E}(Y^n) = \frac{n+(n+4)\,3^n}{(n+1)\,4^{n+1}}} \qquad (n \geq 1).$$

Comparing with X

For the uniform variable, $\mathrm{E}(X^n) = \int_0^1 x^n \,\mathrm{d}x = \dfrac{1}{n+1}$.

With $n=1$, the formula gives $\mathrm{E}(Y) = \dfrac{1+5\cdot 3}{2\cdot 16} = \dfrac{1}{2} = \mathrm{E}(X)$, as required. (This is no surprise: the clipping moves the mass at the two ends inwards by matching amounts, symmetrically about $\frac{1}{2}$.)

With $n=2$, $\mathrm{E}(Y^2) = \dfrac{2+6\cdot 9}{3\cdot 64} = \dfrac{56}{192} = \dfrac{7}{24}$, while $\mathrm{E}(X^2) = \frac{1}{3} = \frac{8}{24}$. Hence

$$\mathrm{E}(X^2) - \mathrm{E}(Y^2) = \frac{8}{24} - \frac{7}{24} = \frac{1}{24},$$

as required.

For general $n \geq 2$, write both moments over the common denominator $(n+1)4^{n+1}$:

$$\mathrm{E}(X^n) = \frac{4^{n+1}}{(n+1)4^{n+1}}.$$

So we must show that $4^{n+1} > n+(n+4)3^n$. Using $4^{n+1} = 4(3+1)^n$ and the binomial theorem, and keeping the top two terms separate,

$$4^{n+1} = 4\cdot 3^n + 4n\cdot 3^{n-1} + 4\sum_{k=0}^{n-2}\binom{n}{k}3^k.$$

Subtracting $n+(n+4)3^n = 4\cdot 3^n + 3n\cdot 3^{n-1} + n$ gives

$$4^{n+1} - (n+(n+4)3^n) = n\,3^{n-1} - n + 4\sum_{k=0}^{n-2}\binom{n}{k}3^k.$$

Ask yourself Why is this difference positive?

For $n \geq 2$ we have $3^{n-1} \geq 3$, so $n\,3^{n-1} - n \geq 2n > 0$. The sum is also positive, since it contains the term $k = 0$, which equals 1, and every term is non-negative. Hence the difference is strictly positive, and

$$\mathrm{E}(X^n) > \mathrm{E}(Y^n) \qquad \text{for all } n \geq 2.$$

(For $n = 1$ the difference is $3^0 - 1 + 0 = 0$, which matches $\mathrm{E}(X) = \mathrm{E}(Y)$.)

The sum of 240 000 copies

From the previous part,

$$\mathrm{Var}(Y) = \mathrm{E}(Y^2) - (\mathrm{E}(Y))^2 = \frac{7}{24} - \frac{1}{4} = \frac{1}{24}.$$

Let $S = Y_1 + \cdots + Y_N$ with $N = 240\,000$. Since the Y_i are independent, expectations and variances add:

$$\mathrm{E}(S) = \frac{N}{2} = 120\,000, \qquad \mathrm{Var}(S) = \frac{N}{24} = 10\,000,$$

so S has standard deviation 100. The Y_i are independent and identically distributed with finite variance and N is huge, so by the central limit theorem S is approximately $\mathrm{N}(120\,000,\, 100^2)$. Hence

$$\mathrm{P}(S < K) \approx \Phi\Big(\frac{K - 120\,000}{100}\Big) = \frac{3}{4} \quad \Longrightarrow \quad \frac{K - 120\,000}{100} = \Phi^{-1}(0.75) \approx 0.6745,$$

so

$$K \approx 120\,000 + 67.45 \approx 120\,067.$$

No continuity correction is needed: Y is not integer-valued, and with a standard deviation of 100 any such correction would make no difference to this accuracy anyway.

Checks

- The formula for $\mathrm{E}(Y^n)$ lies between $(\frac{1}{4})^n$ and $(\frac{3}{4})^n$, as it must, since $\frac{1}{4} \leq Y \leq \frac{3}{4}$. For $n = 3$ it gives $\frac{3+7\cdot 27}{4\cdot 256} = \frac{192}{1024} = \frac{3}{16} = 0.1875$, and a midpoint-rule integral of $g(x)^3$ agrees to ten decimal places.
- $\mathrm{Var}(Y) = \frac{1}{24} < \frac{1}{12} = \mathrm{Var}(X)$: clipping pulls values towards the mean, so the spread should drop.
- A simulation of 20 000 sums of 2400 copies of Y gave a standardised upper quartile of about 0.678, close to 0.6745, which supports the normal approximation.

Common slips

- Treating Y as uniform on $[\frac{1}{4}, \frac{3}{4}]$, and so forgetting the point masses of $\frac{1}{4}$ at each end. A uniform variable on that interval has a density of 2, not 1.

- Using $\mathrm{Var}(S) = N^2\,\mathrm{Var}(Y)$ rather than $N\,\mathrm{Var}(Y)$. Independence makes variances add, so the standard deviation grows like $\sqrt{N}$.
- Getting the sign wrong in the last step. $\mathrm{P}(S < K) = \frac{3}{4} > \frac{1}{2}$, so K must be above the mean.

Beyond STEP

Y is a clipped (or "Winsorised") version of X, and it has a distribution that is neither discrete nor continuous. Measure-theoretic probability handles such distributions uniformly through the distribution function, and every expectation becomes a single integral $\int g\,\mathrm{d}F$. The inequality $\mathrm{E}(X^n) > \mathrm{E}(Y^n)$ with equal means is an instance of convex ordering: X is a "mean-preserving spread" of Y, so $\mathrm{E}(\phi(X)) \geq \mathrm{E}(\phi(Y))$ for every convex ϕ. The central limit theorem used at the end is proved in Grimmett and Welsh, *Probability: An Introduction*.

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